Yes, this is a very small dataset. With ~7 features and 2000 rows, neural networks are usually **unnecessary** and often worse than simpler models.

What a “small dataset” means in practice

Size	Example	When it makes sense
Very small	2–5k rows, a few to a dozen features	Almost always classical ML
Small	10–50k rows	Usually still better: trees / boosting
Medium	100k+ rows, complex data (images, text)	Neural networks start to make sense
Large	Millions of rows / large models	Neural networks (or larger)

Your 2000 rows × 7 features are a **very small** dataset.

Is it worth using neural networks?

Probably not. At this size:

- Neural networks **overfit** easily.
- They are slower to train and harder to interpret.
- Simpler models usually give **better or comparable** results.

Better options to start with:

Model	Why it fits
Random Forest	Robust against overfitting, little tuning
XGBoost / LightGBM	Often the best result on tabular data
Logistic Regression / Linear	Simple, fast, interpretable
SVM	Good for smaller datasets

When to use a neural network anyway?

- When the data are **non-linear and complex** and classical models perform poorly.
- When you want to **learn** neural networks (then go ahead, but the result will not necessarily be better).
- When you plan a much larger dataset later.

Summary

- 2000 rows + 7 features = **very small data**.
- You can use neural networks, but you **don’t have to**, and they will probably not give a better result than XGBoost or Random Forest.
- On such a dataset you can easily train a model even on a weak computer (no GPU).

Would you like me to show a simple example in Python (e.g. XGBoost or a small neural network) on data from a CSV file? Or help you decide which model to choose for your task (classification or regression)?

Roughly how many epochs do you think I should set? I’m using early stopping.

With early stopping you don’t have to guess the exact number of epochs. Instead you set an upper limit, and early stopping stops training by itself once the model stops improving.

Recommendation for your case (approx. 2000 rows, 7 features)

Parameter	Suggested value	Comment
Maximum number of epochs	**100–300**	Safe upper limit
Patience (early stopping)	**10–20**	How many epochs to wait without improvement
Monitor	val_loss	Usually the best choice

Example settings

- epochs=200
- patience=15
- min_delta=0.001 (optional — the minimum improvement that counts)

Why?

- On a small dataset the network usually converges quickly (often within 20–80 epochs).
- Early stopping will stop training earlier anyway, so a high limit does no harm.
- Too low a limit (e.g. 20–30) can stop training before the optimum is reached.
- Too high a limit (e.g. 1000+) brings nothing, because early stopping stops earlier anyway.

Additional tips

- Always use a validation set (e.g. 15–20% of the data).
- If the model reaches a plateau very quickly — you can lower the learning rate.
- With such a small dataset, make sure the model does not overfit (regularisation, dropout, a smaller network).

Summary: Set **200 epochs** with **patience=15**. Early stopping will decide by itself when to finish.